Brhanu Fentaw Znabu

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Research Interests

My research develops deep learning foundation and generative models for large-scale biological sequence analysis across genomes and proteins, with a focus on domain-adaptive pre-training, generative design, and rigorous leakage-aware evaluation. I build systems end to end, from data curation through HPC training to open-source release, and apply them to predicting evolutionary trajectories and protein function.

Education

Ph.D. Biomedical Engineering — Aug 2024 – Expected May 2029
University of Nebraska–Lincoln — *Lincoln, NE*

- Advisor: Nicole R. Sexton, PhD | Nebraska Center for Virology

M.Sc. Biomedical Science and Engineering — 2021 – 2024
Gwangju Institute of Science and Technology (GIST) — *South Korea*

- GPA: 4.0/4.5 | Advisor: Euiheon Chung, PhD
- Thesis: Deep behavioral phenotyping of diabetic neuropathy mouse model using open-field test

B.Sc. Biomedical Engineering — 2014 – 2018
Jimma University — *Ethiopia*

- GPA: 3.64/4.0 | Graduated with Distinction (Ranked 4th of 180)

Research Experience

PhD Researcher — Aug 2024 – Present
Sexton Lab, University of Nebraska–Lincoln — *Lincoln, NE*

- Built ArboFM, a domain-adapted transformer model via continued masked language model pre-training of DNABERT-2 (117M params) on 120K+ sequence windows across 362 species; achieved AP = 0.978 under species-grouped cross-validation preventing data leakage.
- Designed a leakage-aware evaluation framework quantifying 15-pp performance inflation from phylogenetic data leakage in cross-validation; applied gradient-based attribution and 6-mer enrichment analysis for model interpretability.
- Built a cross-domain transfer learning matrix across four biological families (6,052 sequences, 124K+ windows); demonstrated that learned representations are domain-specific (Jaccard ≈ 0), informing transfer learning strategy design.
- Trained species-stratified classifiers (logistic regression, random forest) on 1,285 sequences with 97 engineered features; PR-AUC = 1.000 for binary classification with CpG suppression as top feature via feature importance analysis.
- Performed retrospective temporal validation demonstrating the model detects distribution shifts in held-out future data, including within-class lineage discrimination (CLES = 0.898).
- Developed ViraPredict, a multi-modal forecasting framework integrating fitness landscape modeling, ESM-2 (650M params) zero-shot mutation scoring (F1 = 0.936), and LoRA-fine-tuned DNABERT-2 with temporal embeddings on 29,493 curated sequences across 130 countries.

Research Assistant — Mar 2021 – Jun 2024
Neurophotonics Lab, GIST — *South Korea*

- Developed unsupervised autoregressive and Hidden Markov models for temporal sequence modeling and state transition detection in behavioral time-series data.
- Benchmarked deep learning-based 3D pose estimation (DeepLabCut) against traditional 2D analysis using logistic regression on multi-dimensional behavioral features.

Open-Source Projects

- **ProtDiff-ESM** (github.com/brhanufen/protdiff-esm): Discrete-diffusion protein language model conditioned on frozen ESM-2 with classifier-free guidance toward a fitness target; training and reverse-diffusion sampler built from scratch, evaluated on ProteinGym with a leakage-aware framework.
- **AbStab** (github.com/brhanufen/abstab): Leakage-aware benchmark for antibody thermostability prediction; showed random splits overstate generalization to novel antibodies and that a general PLM (ESM-2) relies on clonal leakage more than an antibody-specific model (AntiBERTy).
- **PerturbVAE** (github.com/brhanufen/perturbvae): Conditional VAE for single-cell perturbation-response prediction, evaluated under random, function-grouped, and combinatorial held-out splits on Norman 2019 and Replogle 2022 Perturb-seq data.

Publications

1. Ashiquzzaman A*, Lee E*, **Znabu BF***, Sakib AN, Chung G, Kim SS, Kim YR, Kwon H-S, Chung E. MoSeq based 3D behavioral profiling uncovers neuropathic behavior changes in diabetic mouse model. *Scientific Reports* 15, 15114 (2025)
2. **Znabu BF**, Atif Z. Interpretable deep learning-based multi-omics integration for prognosis in hepatocellular carcinoma. *bioRxiv*. 2026. DOI: 10.64898/2026.04.01.715980v1 (Preprint)
3. **Znabu BF**, Sexton NR. A genomic foundation model for predicting arbovirus epidemic emergence across RNA virus families. *Bioinformatics* (In preparation)
4. **Znabu BF**, Sexton NR. Deciphering the host-switching grammar of flaviviruses using foundation model embeddings: a leakage-aware evaluation framework. *PLOS Computational Biology* (In preparation)
5. **Znabu BF**, Sexton NR. Cross-family comparison of host-switching grammar in arthropod-borne viruses reveals convergent evolution of family-specific immune evasion strategies. *Virus Evolution* (In preparation)
6. **Znabu BF**, Sexton NR. Genome-scale classification of flavivirus host range using composition and codon-usage signatures. *Journal of Virology* (In preparation)

Technical Skills

Programming: Python, R, C++, MATLAB | **ML/DL:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, DNABERT-2, ESM-2, AntiBERTy | **Infrastructure:** GPU (V100), HPC (Slurm), Git/GitHub

Methods: Foundation model pre-training (MLM), discrete diffusion, classifier-free guidance, variational autoencoders, LoRA fine-tuning, transfer learning, leakage-aware evaluation, gradient attribution, zero-shot prediction, PCA, UMAP | **Data:** NCBI GenBank, GISAID, ProteinGym, TCGA, GEO, Biopython, ESMFold, IEDB

Academic Service

Poster Judge — April 16, 2026
School of Biological Sciences Undergraduate Research Symposium, UNL — *Lincoln, NE*

Teaching Experience

Lecturer — Aug 2019 – Jan 2021
Hawassa University — *Ethiopia*

- Taught Biomechanics, Biomaterials, and Biophysics to undergraduate biomedical engineering students.

Awards & Grants

- Nominated by UNL as one of four students for the Google PhD Fellowship 2026 in AI for Health
- Korean Government Scholarship for Master's Study, GIST (2020)
- Research Grant for Colostomy Device Development, Hawassa University (2020)
- Best B.Sc. Thesis Award, Ethiopian Science, Technology, and Innovation (2018)